Worksheet 4.4 — Full Curve Sketching

Grade 12 Calculus and Vectors (MCV4U) · Unit 4: Curve Sketching

The Big Ideas

Run the checklist: domain/symmetry, intercepts, asymptotes, f' (extrema), f'' (concavity/inflection), end behaviour then sketch.

- Domain, symmetry, intercepts first.
- f' extrema; f'' inflection/concavity.
- Finish with end behaviour and a consistent sketch.

The Big Picture

A systematic method

Curve sketching combines the whole toolkit: domain, intercepts, asymptotes, f' for increase/decrease, and f'' for concavity.

Assemble the evidence

Mark critical points, inflection points, and asymptotes, then record the sign of f' and f'' on each interval.

Then draw

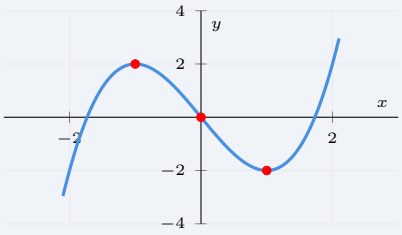
Only after gathering every feature do you sketch the graph is the summary, not the starting point.

Worked Examples

Example 1 — Cubic

Sketch $f(x) = x^3 - 3x$.

Solution. **Step 1 (intercepts):** $f(x) = x(x^2 - 3) = 0 \Rightarrow x = 0, \pm\sqrt{3}$; y-intercept $f(0) = 0$.
Step 2 (extrema from f'): $f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$; sign test gives max $(-1, 2)$, min $(1, -2)$.
Step 3 (inflection from f''): $f''(x) = 6x = 0 \Rightarrow x = 0$, so $(0, 0)$.
Step 4 (end behaviour): odd degree, positive leading coefficient $\Rightarrow$ down–up.
Conclusion: x-intercepts $0, \pm\sqrt{3}$; max $(-1, 2)$, min $(1, -2)$; inflection $(0, 0)$; ends down–up:



Example 2 — Quartic (even)

Sketch $f(x) = x^4 - 2x^2$.

Solution. **Step 1 (symmetry):** only even powers of $x \Rightarrow$ even function, symmetric about the y-axis.

Step 2 (intercepts): $x^2(x^2-2) = 0 \Rightarrow x = 0, \pm\sqrt{2}$.

Step 3 (extrema from f'): $f'(x) = 4x^3 - 4x = 4x(x-1)(x+1) = 0 \Rightarrow x = 0, \pm 1$; testing signs gives max $(0,0)$, mins $(\pm 1, -1)$.

Step 4 (inflection from f''): $f''(x) = 12x^2 - 4 = 0 \Rightarrow x = \pm\frac{1}{\sqrt{3}}$.

Step 5 (end behaviour): even degree, positive leading coefficient $\Rightarrow$ up–up.

Conclusion: x-intercepts $0, \pm\sqrt{2}$; max $(0,0)$, mins $(\pm 1, -1)$; inflections at $x = \pm\frac{1}{\sqrt{3}}$; ends up–up (a "W" shape).

Example 3 — Cubic

Sketch $f(x) = x^3 - 3x^2$.

Solution. **Step 1 (intercepts):** $x^2(x-3) = 0 \Rightarrow x = 0, 3$.

Step 2 (extrema from f'): $f'(x) = 3x^2 - 6x = 3x(x-2) = 0 \Rightarrow x = 0, 2$; testing signs gives max $(0,0)$, min $(2,-4)$.

Step 3 (inflection from f''): $f''(x) = 6x - 6 = 0 \Rightarrow x = 1$, so $(1,-2)$.

Step 4 (end behaviour): odd degree, positive leading coefficient $\Rightarrow$ down–up.

Conclusion: x-intercepts $0, 3$; max $(0,0)$, min $(2,-4)$; inflection $(1,-2)$; ends down–up.

Example 4 — From clues

$f' > 0$ on $(-\infty, 0) \cup (2, \infty)$, $f' < 0$ on $(0,2)$, $f'' > 0$ on $(1, \infty)$.

Solution. **Step 1 (locate extrema from the f' sign changes):** $+ \to -$ at $x = 0$ (local max); $- \to +$ at $x = 2$ (local min).

Step 2 (locate the inflection from the f'' sign change): concave up begins at $x = 1$, so that's the inflection point.

Conclusion: max at $x = 0$, min at $x = 2$, inflection at $x = 1$ — consistent with a cubic shape.

Example 5 — Match f to f'

f has a max at -1, min at 1. What is f'?

Solution. **Step 1 (recall what a max/min means for f'):** $f' = 0$ at both $x = -1$ and $x = 1$.

Step 2 (use the sign pattern of a max-then-min): f' must be $+$ before -1, $-$ between -1 and 1, then $+$ again after 1 — a parabola opening upward with roots at ± 1.

Conclusion: f' is an upward-opening parabola with zeros at $x = \pm 1$, e.g. $f'(x) = 3x^2 - 3$.

Example 6 — Intercepts

x-intercepts of $f(x) = x^3 - 3x$.

Solution. **Step 1 (factor):** $f(x) = x(x^2 - 3)$.

Step 2 (set each factor to 0): $x = 0$ or $x^2 = 3 \Rightarrow x = \pm\sqrt{3}$.

Conclusion: x-intercepts at $0, \pm\sqrt{3}$.

Example 7 — Symmetry

Is $f(x) = x^4 - 2x^2$ even or odd?

Solution. **Step 1 (test $f(-x)$):** $f(-x) = (-x)^4 - 2(-x)^2 = x^4 - 2x^2 = f(x)$.

Step 2 (interpret): $f(-x) = f(x)$ is the definition of even.

Conclusion: even (only even powers of x appear).

Example 8 — Extrema

Local extrema of $f(x) = x^3 - 3x^2$.

Solution. **Step 1 (differentiate and find critical numbers):** $f'(x) = 3x^2 - 6x = 3x(x-2) = 0 \Rightarrow x = 0, 2$.

Step 2 (test signs to classify): $+ \to -$ at $x = 0$ (max); $- \to +$ at $x = 2$ (min).

Conclusion: max $(0, 0)$, min $(2, -4)$.

Example 9 — Always increasing

If $f' > 0$ everywhere, what does the curve do?

Solution. **Step 1 (recall the meaning of $f' > 0$):** the slope is positive at every point.

Step 2 (interpret for the whole curve): a curve with positive slope everywhere never turns around.

Conclusion: the graph is always increasing, with no local extrema.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

x-intercepts of $f(x) = x^3 - 3x$?

Question 2

Local extrema of $f(x) = x^3 - 3x^2$?

Question 3

Inflection point of $f(x) = x^3 - 3x$?

Question 4

Is $f(x) = x^4 - 2x^2$ even or odd?

Question 5

If $f' > 0$ everywhere, what does the graph do?

Question 6

y-intercept of $f(x) = x^3 - 3x + 2$?

Question 7

Local extrema of $f(x) = x^3 - 3x$?

Question 8

End behaviour of $f(x) = x^4 - 2x^2$?

Question 9

Inflection point of $f(x) = x^3 - 3x^2$?

Question 10

If f has a min at 2, what does f' do at 2?

Question 11

x-intercepts of $f(x) = x^4 - 2x^2$?

Question 12

Is $f(x) = x^3 - 3x$ even, odd, or neither?

Question 13 — Challenge

Do a full sketch analysis of $f(x) = x^3 - 3x^2$ (intercepts, extrema, inflection, ends).

Question Recap & Answer Key

Question Recap

1. x-intercepts of $f(x) = x^3 - 3x$?
2. Local extrema of $f(x) = x^3 - 3x^2$?
3. Inflection point of $f(x) = x^3 - 3x$?
4. Is $f(x) = x^4 - 2x^2$ even or odd?
5. If $f' > 0$ everywhere, what does the graph do?
6. y-intercept of $f(x) = x^3 - 3x + 2$?
7. Local extrema of $f(x) = x^3 - 3x$?
8. End behaviour of $f(x) = x^4 - 2x^2$?
9. Inflection point of $f(x) = x^3 - 3x^2$?
10. If f has a min at 2, what does f' do at 2?
11. x-intercepts of $f(x) = x^4 - 2x^2$?
12. Is $f(x) = x^3 - 3x$ even, odd, or neither?
13. Do a full sketch analysis of $f(x) = x^3 - 3x^2$ (intercepts, extrema, inflection, ends).

✓ Answer Key

1. $0, \pm\sqrt{3}$
2. max $(0, 0)$, min $(2, -4)$
3. $(0, 0)$
4. even
5. always increasing
6. $(0, 2)$
7. max $(-1, 2)$, min $(1, -2)$
8. upup
9. $(1, -2)$
10. $f' = 0$, changing $- \to +$
11. $0, \pm\sqrt{2}$
12. odd
13. x-ints $0, 3$; max $(0, 0)$, min $(2, -4)$; inflection $(1, -2)$; ends downup